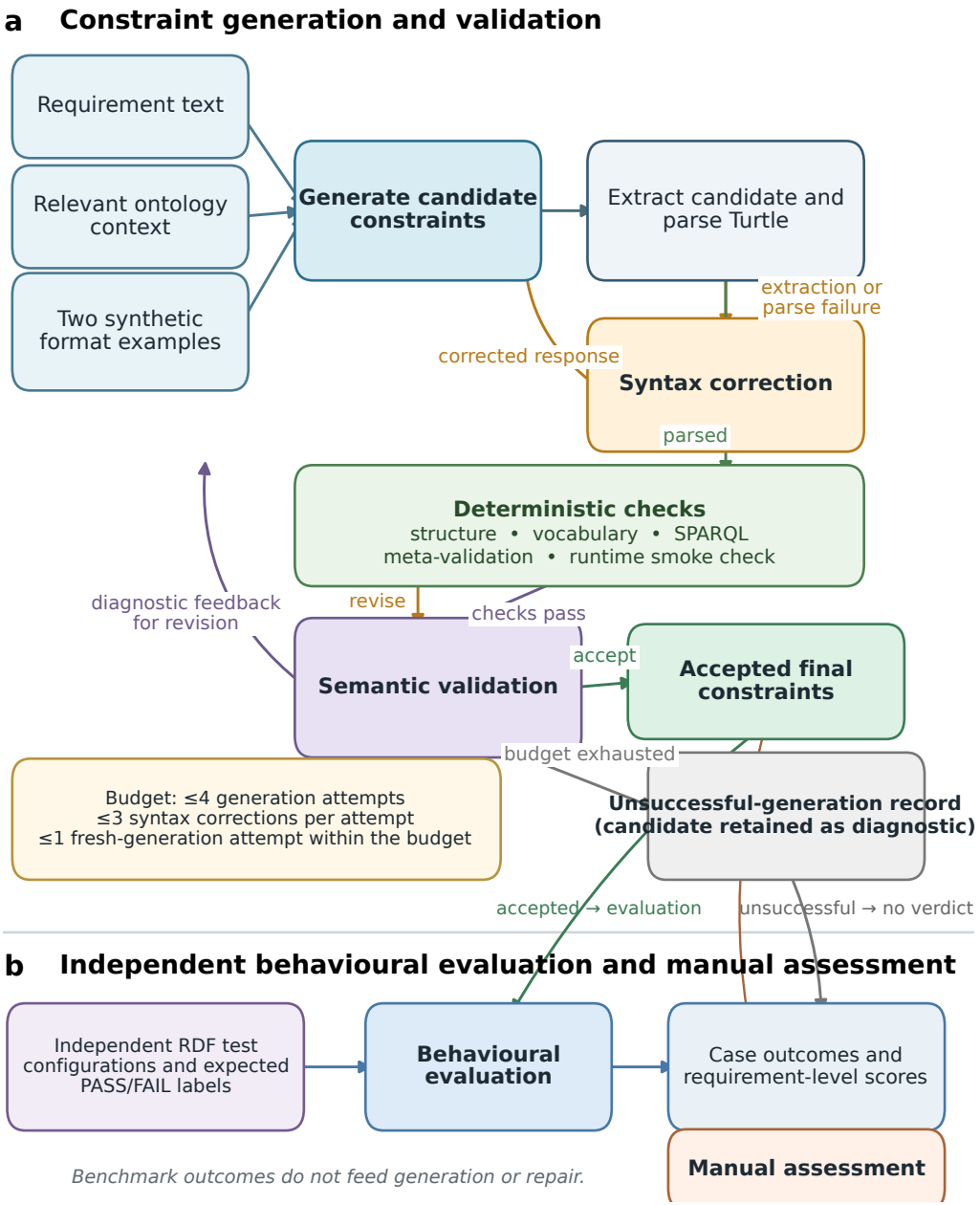
a Constraint generation and validation
Requirement text
Relevant ontology context
Two synthetic format examples
Generate candidate constraints
Extract candidate and parse Turtle
extraction or parse failure
corrected response
Syntax correction
parsed
Deterministic checks
structure • vocabulary • SPARQL
meta-validation • runtime smoke check
diagnostic feedback for revision
revise
checks pass
Semantic validation
accept
Accepted final constraints
budget exhausted
Budget: ≤4 generation attempts
≤3 syntax corrections per attempt
≤1 fresh-generation attempt within the budget
Unsuccessful-generation record
(candidate retained as diagnostic)
accepted → evaluation
unsuccessful → no verdict
b Independent behavioural evaluation and manual assessment
Independent RDF test configurations and expected PASS/FAIL labels
Behavioural evaluation
Case outcomes and requirement-level scores
Manual assessment
Benchmark outcomes do not feed generation or repair.